

STELLAR END POINTS

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THE FORM A STAR TAKES in the ultimate stage of its life is called a stellar end point. Such end points include some of the most exotic objects in the Milky Way. The fate of a star is dictated by its mass, with lower-mass stars becoming white dwarfs and the highest-mass stars becoming black holes from which not even light can escape. Between these are neutron stars, including spinning pulsars.

WHITE DWARFS

Once a star has used up all of its fuel through nuclear fusion, the stellar remnant will collapse, as it cannot maintain enough internal pressure to counteract its gravity. Stars of less than about 8 solar masses will lose up to 90 percent of their material in stellar winds and by creating planetary nebulae (see p.255). If the remnants of these stars have less than 1.4 solar masses (the Chandrasekhar limit), they will become white dwarfs. White dwarfs are supported by what is known as electron degeneracy pressure, created by the repulsion between electrons in their core material. More massive stars collapse to the smallest diameters and highest densities. The first white dwarf to be discovered, Sirius B (see p.268), has a mass similar to that of the Sun but a radius only twice that of the Earth. Although they have surface temperatures of around 180,000°F (100,000°C) at first, white dwarfs fade over periods of hundreds of millions of years, eventually becoming cold black dwarfs.

WHITE DWARFS IN NGC 6791

The faint stars inside the squares in this image are white dwarfs in the globular cluster NGC 6791. Too faint to be seen from the ground, the stars were captured here by the Hubble Space Telescope.

MORGUE OF STARS

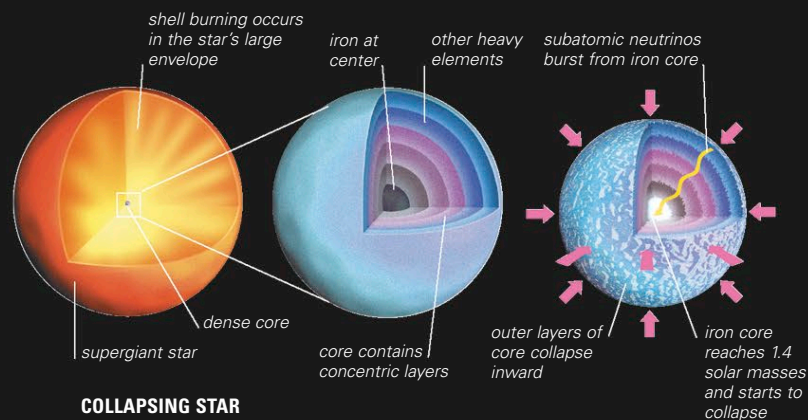
Spanning a distance of 900 light-years, this mosaic of X-ray images of the center of the Milky Way reveals hundreds of white-dwarf stars, neutron stars, and black holes. They are all embedded in a hot, incandescent fog of interstellar gas. The supermassive black hole at the center of the galaxy is located inside the central bright white patch.

SUPERNOVAE

Massive stars die spectacularly, blasting their outer layers off into space in type II supernovae explosions. A type I supernova is a type of variable star (see p.283). When a star of more than about 10 solar masses reaches the end of its hydrogen-burning stage, it will eventually produce an iron core. Initially, this core is held up by its internal pressure, but when it reaches a mass greater than 1.4 solar masses (the Chandrasekhar limit), it starts to collapse, forming an extremely dense core almost entirely made of neutrons. Supernova detonation occurs when the outer layers of the star, which have continued to implode, impact on the rigid core and rebound back into space at speeds of up to 45 million mph (70 million kph). This releases massive amounts of energy, creating a great rise in luminosity that may last for several months, before fading. A supernova remnant consisting of the debris will become a nebula.

ASYMMETRIC EXPLOSION

The uneven shape of hot gas clouds expanding from Supernova 1987A indicates extreme turbulence during the initial explosion.



COLLAPSING STAR

As a massive star collapses, elements heavier than helium are produced in a series of shell-burning layers. Elements heavier than iron cannot be produced in this way, and an iron core may collapse to produce a neutron star.